tf.GradientTape Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/GradientTape

> *Record operations for automatic differentiation.*

Function Signatures

```
tf.GradientTape( persistent=False,
watch_accessed_variables=True )
```

Code Examples

```
tf.GradientTape(
persistent=False, watch_accessed_variables=True
)
```


```
tf.GradientTape(
persistent=False, watch_accessed_variables=True
)
```


```
x = tf.constant(3.0)
with tf.GradientTape() as g:
g.watch(x)
y = x * x
dy_dx = g.gradient(y, x)
print(dy_dx)
tf.Tensor(6.0, shape=(), dtype=float32)
```

```
x = tf.constant(3.0)
```

```
with tf.GradientTape() as g:
```

```
dy_dx = g.gradient(y, x)
```

Documentation

Record operations for automatic differentiation.

View aliases

Main aliases

tf.autodiff.GradientTape

See
Migration guide for
more details.

tf.compat.v1.GradientTape

Used in the notebooks

• Introduction to gradients and automatic differentiation

- Advanced automatic differentiation
- Better performance with tf.function
- TensorFlow basics
- Effective Tensorflow 2
- Deep Convolutional Generative Adversarial Network
- DeepDream
- pix2pix: Image-to-image translation with a conditional GAN
- Neural style transfer
- Custom training: walkthrough

Operations are recorded if they are executed within this context manager and at least one of their inputs is being "watched".

Trainable variables (created by tf.Variable or tf.compat.v1.get_variable, where trainable=True is default in both cases) are automatically watched. Tensors can be manually watched by invoking the watch method on this context manager.

For example, consider the function y = x * x. The gradient at x = 3.0 can be computed as:

GradientTapes can be nested to compute higher-order derivatives. For example,

By default, the resources held by a GradientTape are released as soon as GradientTape.gradient() method is called. To compute multiple gradients over the same computation, create a persistent gradient tape. This allows multiple calls to the gradient() method as resources are released when the tape object is garbage collected. For example:

By default GradientTape will automatically watch any trainable variables that are accessed inside the context. If you want fine grained control over which variables are watched you can disable automatic tracking by passing watch_accessed_variables=False to the tape constructor:

Note that when using models you should ensure that your variables exist when using watch_accessed_variables=False. Otherwise it's quite easy to make your first iteration not have any gradients:

Note that only tensors with real or complex dtypes are differentiable.

Methods

batch_jacobian

View source

Computes and stacks per-example jacobians.

See wikipedia article
for the definition of a Jacobian. This function is essentially an efficient implementation of the following:

tf.stack([self.jacobian(y[i], x[i]) for i in range(x.shape[0])]).

Note that compared to GradientTape.jacobian which computes gradient of each output value w.r.t each input value, this function is useful when target[i,...] is independent of source[j,...] for j != i. This assumption allows more efficient computation as compared to GradientTape.jacobian. The output, as well as intermediate activations, are lower dimensional and avoid a bunch of redundant zeros which would result in the jacobian computation given the independence assumption.

Example usage:

gradient

View source

Computes the gradient using operations recorded in context of this tape.

In addition to Tensors, gradient also supports RaggedTensors. For example,

jacobian

View source

Computes the jacobian using operations recorded in context of this tape.

Seewikipedia
article

for the definition of a Jacobian.

Example usage:

View source

Clears all information stored in this tape.

Equivalent to exiting and reentering the tape context manager with a new tape. For example, the two following code blocks are equivalent:

This is useful if you don't want to exit the context manager for the tape, or can't because the desired reset point is inside a control flow construct:

stop_recording

View source

Temporarily stops recording operations on this tape.

Operations executed while this context manager is active will not be recorded on the tape. This is useful for reducing the memory used by tracing all computations.

For example:

View source

Ensures that tensor is being traced by this tape.

watched_variables

View source

Returns variables watched by this tape in order of construction.

__enter__

View source

Enters a context inside which operations are recorded on this tape.

__exit__

View source

Exits the recording context, no further operations are traced.
